

TASK 2: STRUCTURAL SIMULATION (FEA)

Component:

Beam

Software:

PTC Creo Parametric (Simulation)

Material:

Steel_Cast

1. OBJECTIVE

To perform Finite Element Analysis (FEA) on a beam model using different loading and boundary conditions to evaluate:

- Stress distribution
- Deformation
- Factor of Safety

2. MODEL DESCRIPTION

A beam with rectangular cutouts was modeled in Creo Parametric and analyzed using Finite Element Analysis.

3. MATERIAL PROPERTIES (Steel_Cast)

Property	Value
Density	7.85e-06 kg/mm ³
Young’s Modulus	2.0e+05 MPa
Poisson’s Ratio	0.28
Yield Strength	250 MPa
Ultimate Tensile Strength	460 MPa

4. LOAD CASE 1 – CENTER POINT LOAD

Load Applied

1000 N downward load at center.

Boundary Condition

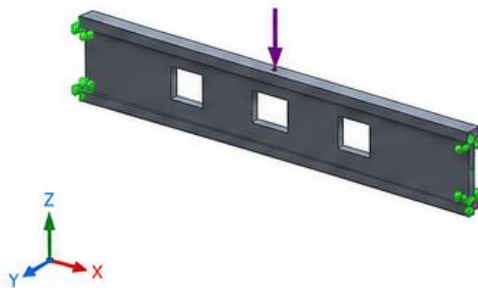
Both ends fixed.

4.1 Geometry and Boundary Condition

3. LOAD CASE 1 – CENTER POINT LOAD

Load: 1000 N downward at the center of the top surface

Boundary Condition: Fixed at both ends
(all DOF constrained)

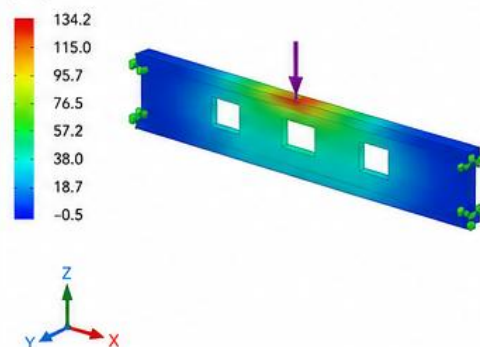


4.2 Von Mises Stress Result

Maximum Stress = **134.2 MPa**

4. LOAD CASE 1 – VON MISES STRESS (MPa)

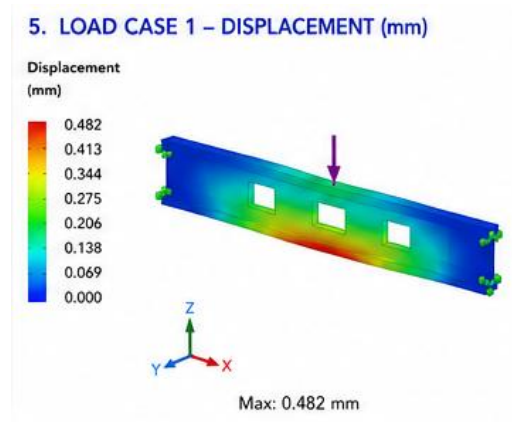
Von Mises Stress
(MPa)



Max: 134.2 MPa

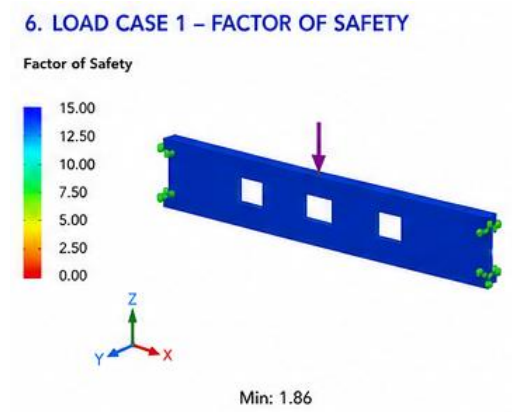
4.3 Displacement Result

Maximum Displacement = **0.482 mm**



4.4 Factor of Safety

Minimum FOS = **1.86**



Observations

- Maximum stress is below yield strength.
- Beam is safe under center point loading.
- Maximum deformation occurs at beam center.

5. LOAD CASE 2 – UNIFORM DISTRIBUTED LOAD

Load Applied

500 N/m uniformly distributed downward load.

Boundary Condition

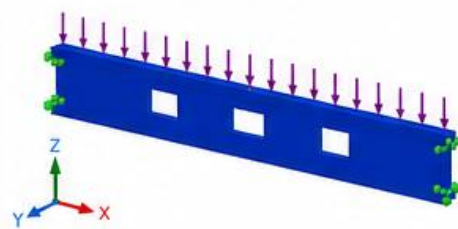
Both ends fixed.

5.1 Geometry and Boundary Condition

7. LOAD CASE 2 – UNIFORM DISTRIBUTED LOAD

Load: Uniformly distributed load of 500 N/m downward

Boundary Condition: Fixed at both ends
(all DOF constrained)



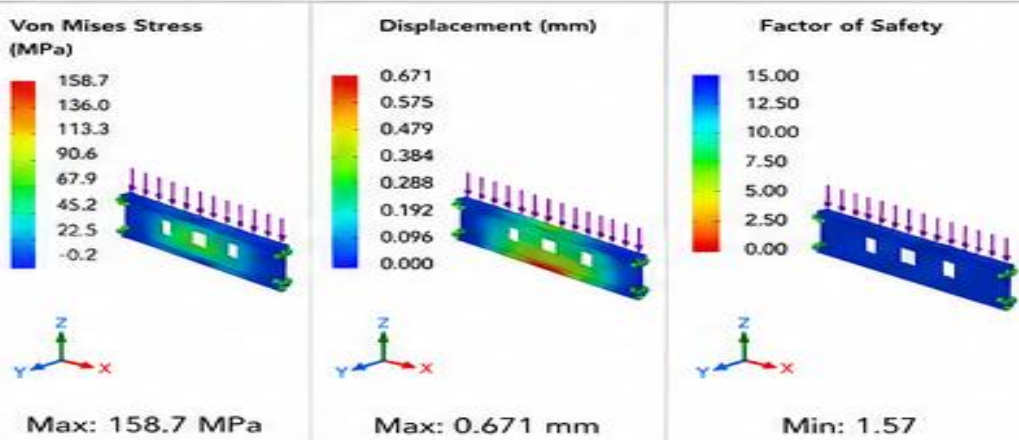
5.2 Von Mises Stress, Displacement, Factor of Safety Result

Maximum Stress = **158.7 MPa**

Maximum Displacement = **0.671 mm**

Minimum FOS = **1.57**

8. LOAD CASE 2 – RESULTS



Observations

- Stress increased under distributed loading.
- Deformation increased compared to Load Case 1.
- Beam remains within safe limits.

6. LOAD CASE 3 – END POINT LOAD (CANTILEVER)

Load Applied

1000 N downward load at free end.

Boundary Condition

One end fixed.

6.1 Geometry and Boundary Condition

9. LOAD CASE 3 – END POINT LOAD (CANTILEVER)

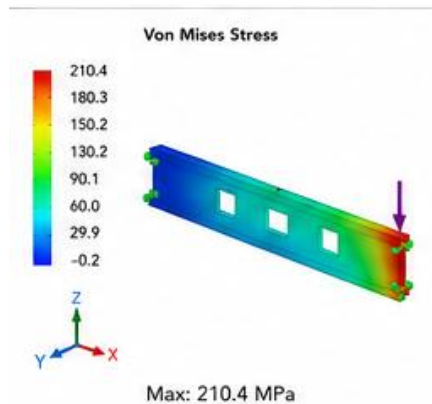
Load: 1000 N downward at the free end

Boundary Condition: Fixed at one end
(cantilever condition)



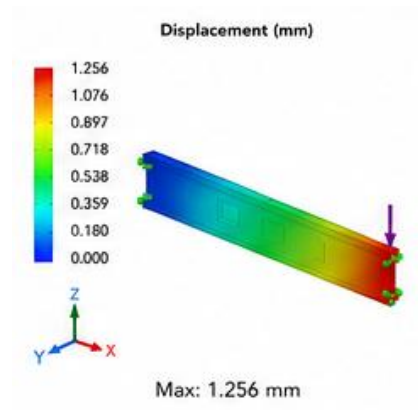
6.2 Von Mises Stress Result

Maximum Stress = **210.4 MPa**



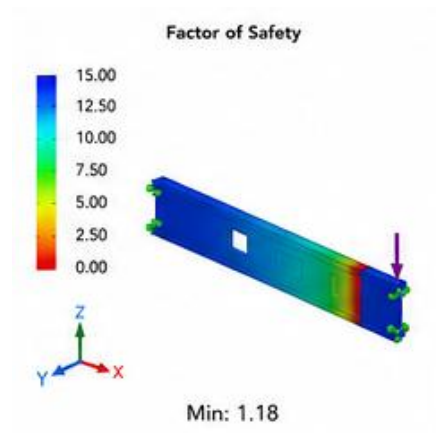
6.3 Displacement Result

Maximum Displacement = **1.256 mm**



6.4 Factor of Safety

Minimum FOS = **1.18**



Observations

- Highest stress occurred in cantilever condition.
- Maximum deformation observed at free end.
- Lowest factor of safety among all cases.

7. SUMMARY TABLE

Load Case	Max Stress (MPa)	Max Displacement (mm)	Min FOS	Status
Case 1	134.2	0.482	1.86	Safe
Case 2	158.7	0.671	1.57	Safe
Case 3	210.4	1.256	1.18	Safe (Low FOS)

8. CONCLUSION

The FEA simulation shows that the beam is safe under all loading conditions because the maximum stress values remain below the material yield strength of Steel_Cast.

Load Case 3 produced:

- Highest stress
- Maximum deformation
- Lowest factor of safety

Therefore, **Load Case 3** is the **critical loading** condition.